InternSync — Diagram pack

Requirements and design diagrams · v1.1 · consistent with SRS v1.1, the SPMP v1.1 and the design document v1.0

No.	Diagram	What it shows	Where to use it
1	Use case diagram (p. 2)	Actors, use cases UC-01 to UC-15, include / extend relations	SRS 3.3 (Table 3.1)
2	Context diagram (p. 3)	System boundary, users and external services	SRS 3.1 (Figure 3.1)
3a	Activity diagram — setup and placement (p. 4)	Swimlanes from opening a cycle to a confirmed placement	SRS 3.2, design document
3b	Activity diagram — training and grading (p. 5)	Weekly loop, final evaluation, grade approval	SRS 3.2, design document
4a, 4b	State machines — report and grade (p. 6)	Report slot and grade life cycles	SRS 4.9 (Figures 4.1, 4.2); design 5.1
4c, 4d	State machines — vacancy and application (p. 7)	Vacancy and application / placement life cycles	SRS 4.4; design 5.1
5	System architecture (p. 8)	Layers, API modules, AI services, scheduler, data	Design 2.1, presentation
6a	ERD — accounts, configuration, profiles, vacancies, matching (p. 9)	Tables, keys and cardinalities (crow's foot)	Design 4.2
6b	ERD — reports, similarity, attendance, evaluations, grades (p. 10)	Tables, keys and cardinalities (crow's foot)	Design 4.2
6c	ERD — platform services (p. 11)	Sessions, tokens, files, notifications, audit, imports, settings, jobs	Design 4.2
7a	Sequence — CV upload and matching (p. 12)	Skill extraction, confirmation, ranking with fallback	Design 6
7b	Sequence — weekly report and review (p. 13)	Submission, Late flag, similarity check, accept / revision	Design 5.1
7c	Sequence — grade submission and approval (p. 14)	Preconditions, approve / return, notifications	Design 5.1
7d	Sequence — scheduled reminders and escalation (p. 15)	15-minute jobs, idempotent notifications	Design 5.3
8	AI pipelines (p. 16)	Fit-score formula, explanation, similarity method	Design 6, presentation
9	Deployment (p. 17)	Zero-cost reference deployment (chosen, CR-01) and the self-hosted option	Design 3.2
10	Schedule (Gantt) (p. 18)	Activities T1–T9, milestones M1–M7, dependencies	SPMP 4, progress presentations
11	Requirements by area and priority (p. 19)	116 requirements by area: Must / Should / Could	SRS 9.2

Colour key: blue = system behaviour or people; orange = AI-supported (advisory) behaviour; green = an approved / final state or the chosen option; grey dashed = external service.
v1.1 (Oct 13, 2026): the ERD is redrawn in three parts to match the data dictionary of the design document (45 tables); the deployment is decided (zero cost, CR-01); UC-15 is now in SRS Table 3.1. Diagram 8 values are configurable defaults tuned on the labelled test sets. Editable sources (Python, Graphviz, Mermaid) are in the source folder; SVG files are in the svg folder.

Figure 1 — Use case diagram

Actors, use cases UC-01 to UC-15, include / extend relations. Use in: SRS 3.3 (Table 3.1).

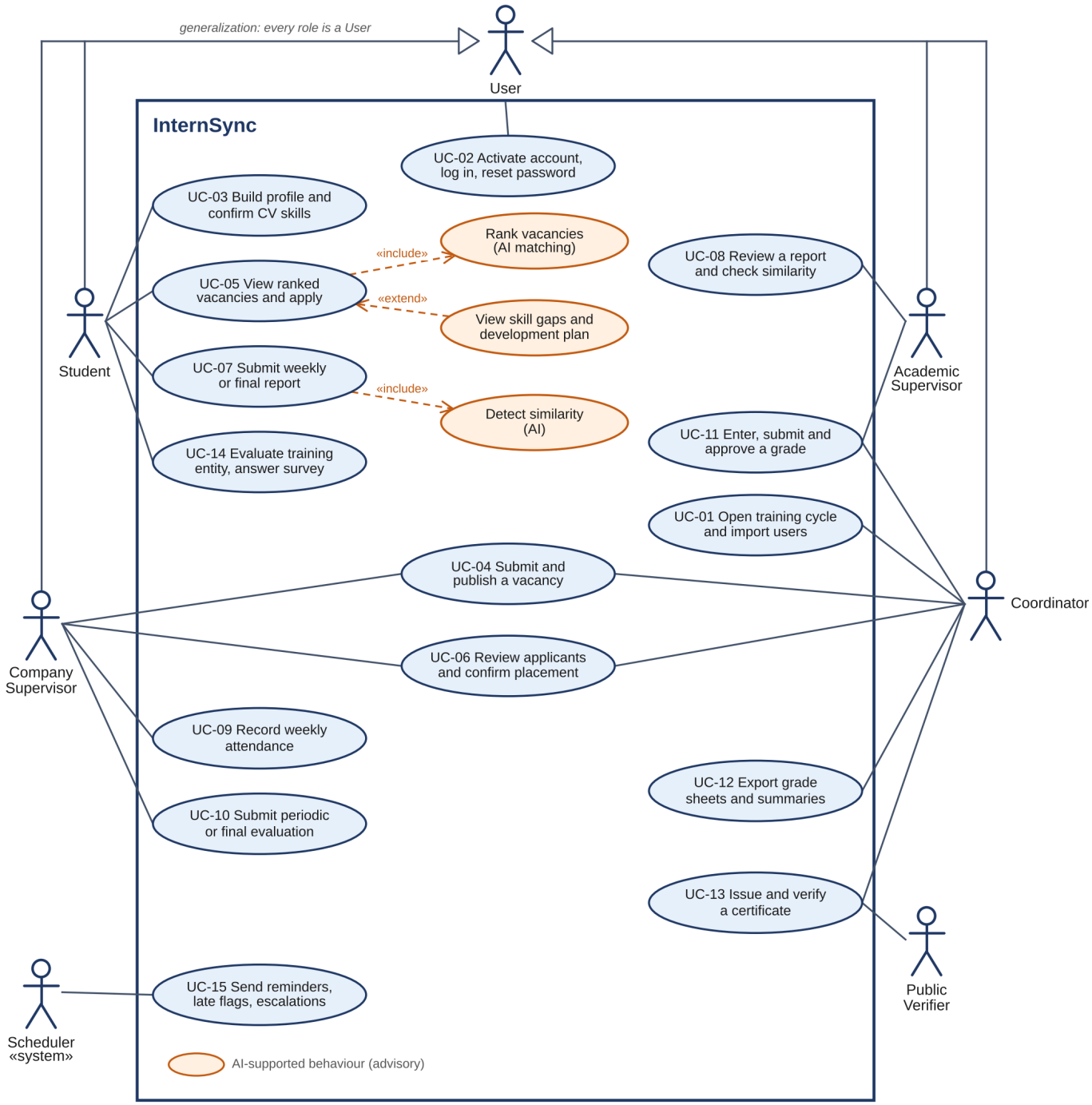


Figure 2 — Context diagram

System boundary, users and external services. Use in: SRS 3.1 (Figure 3.1).

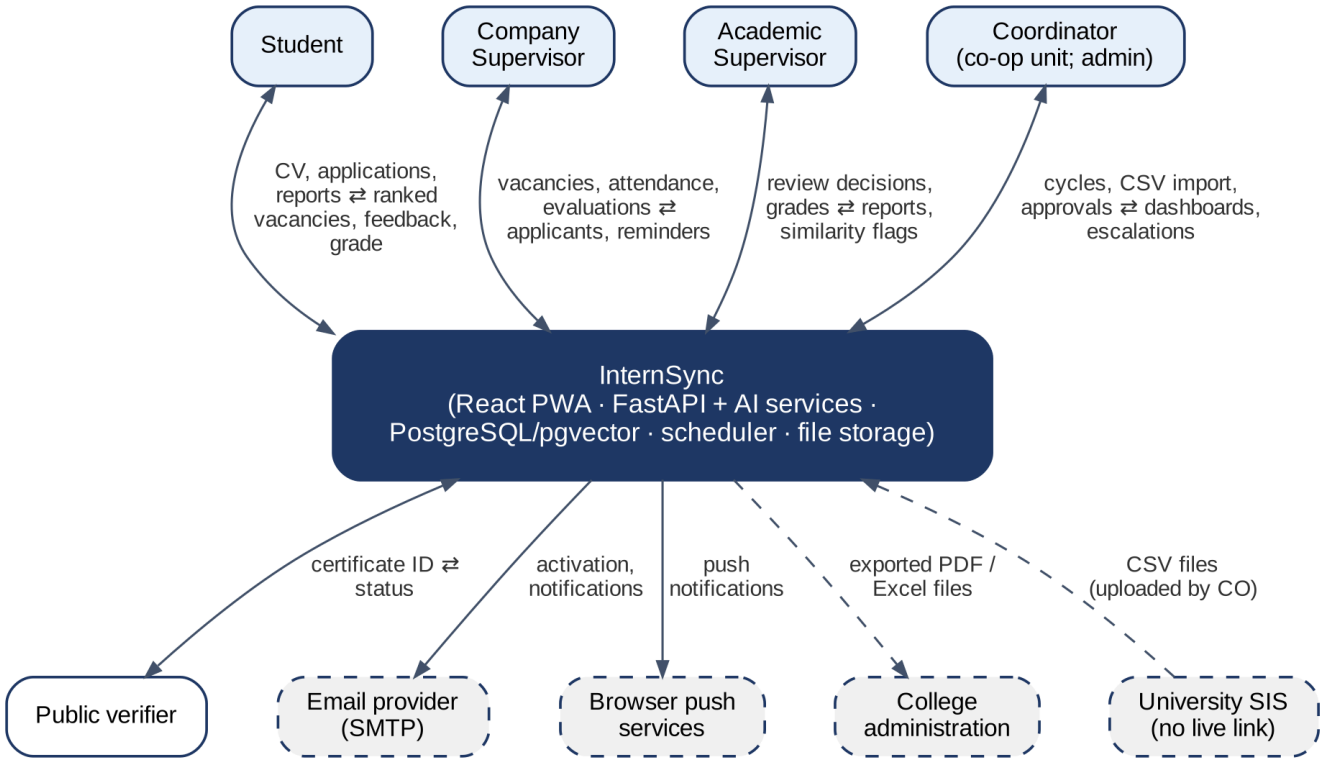


Figure 3a — Activity diagram — setup and placement

Swimlanes from opening a cycle to a confirmed placement. Use in: SRS 3.2, design document.

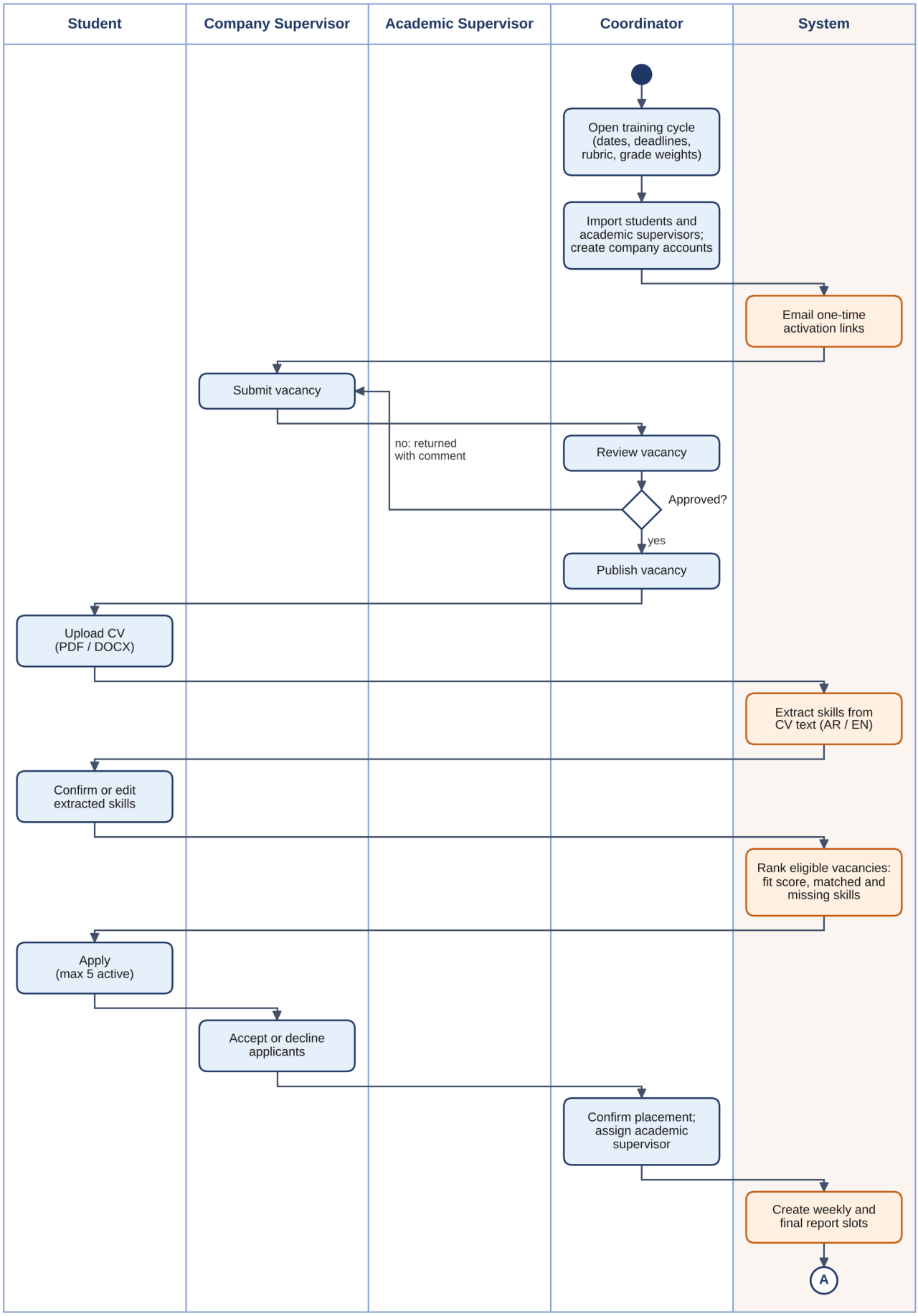


Figure 3b — Activity diagram — training and grading

Weekly loop, final evaluation, grade approval. Use in: SRS 3.2, design document.

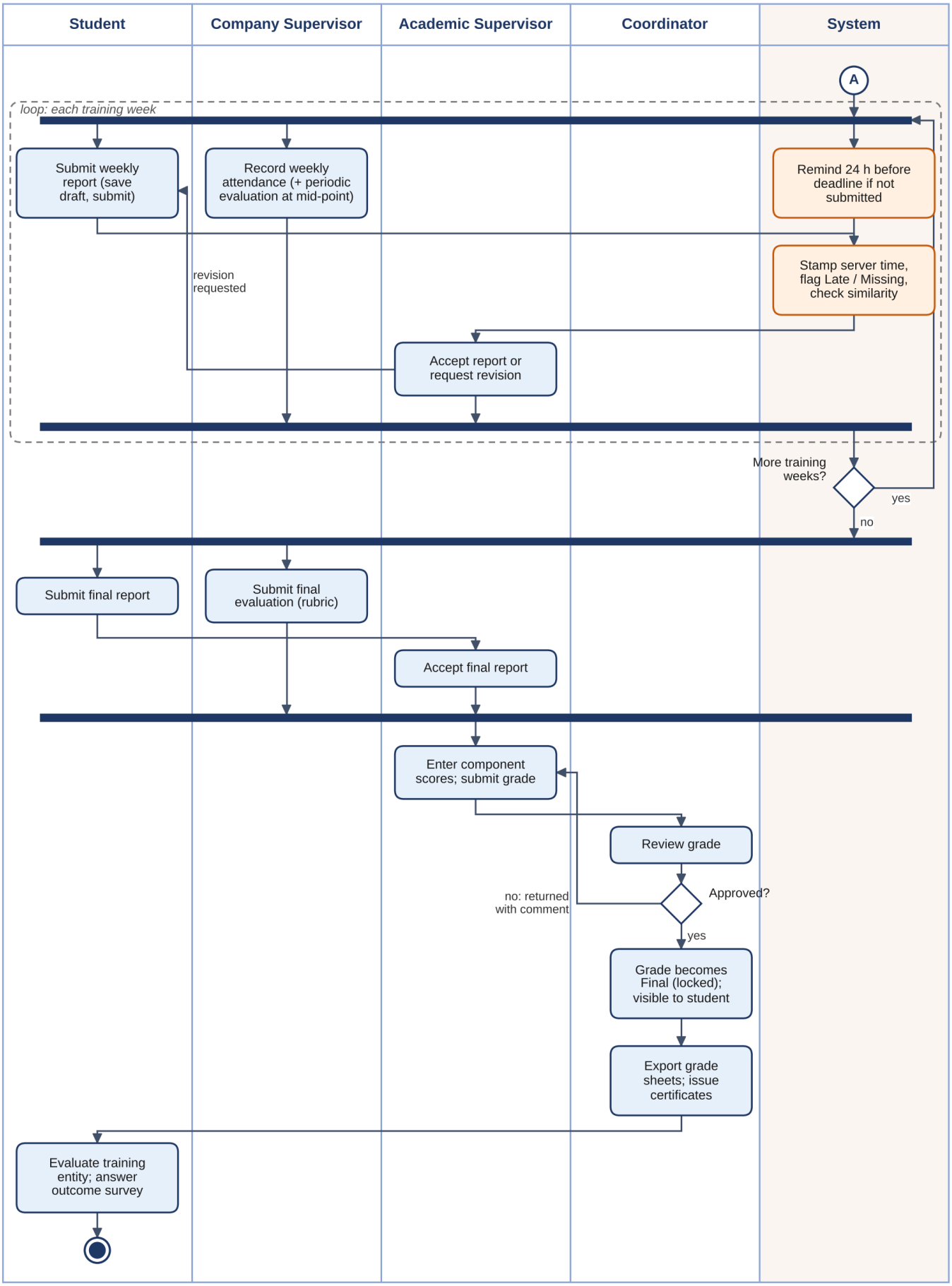
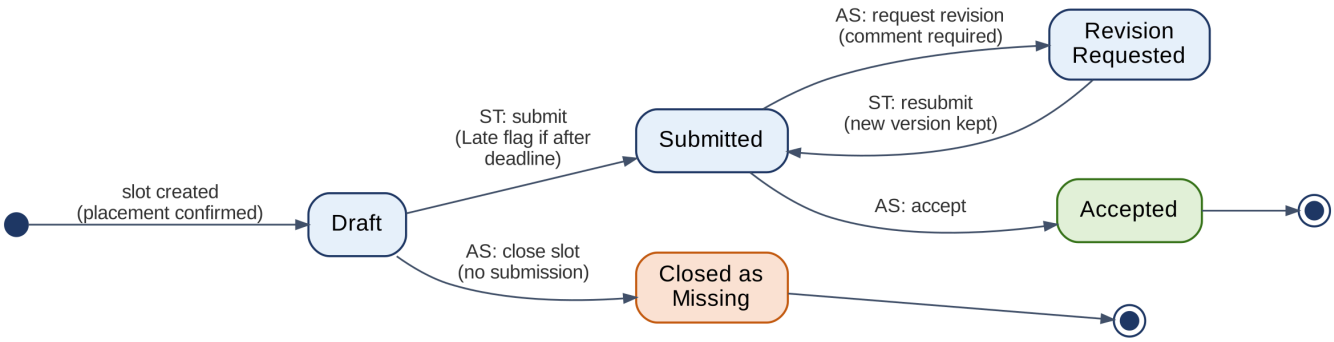


Figure 4a, 4b — State machines — report and grade

Report slot and grade life cycles. Use in: SRS 4.9 (Figures 4.1, 4.2); design 5.1.



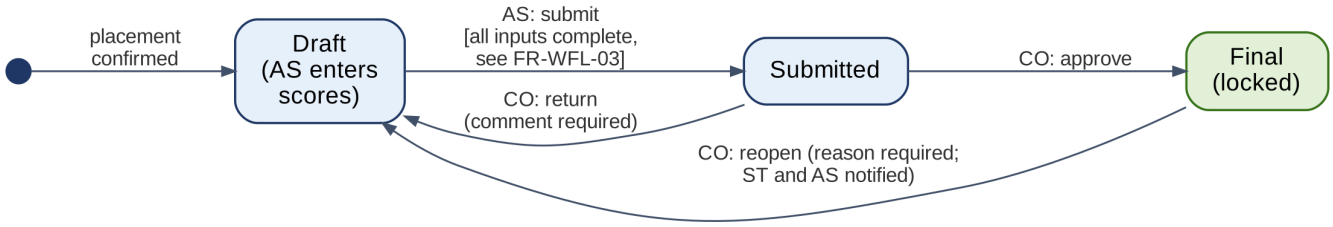
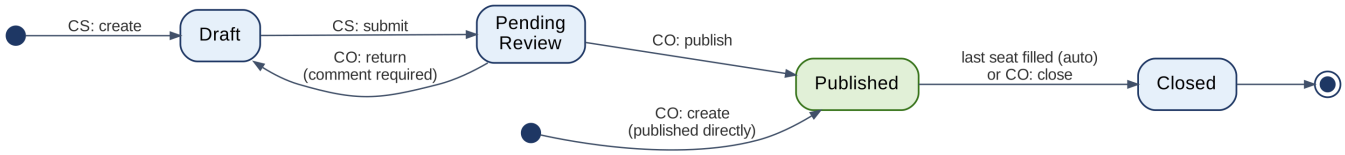


Figure 4c, 4d — State machines — vacancy and application

Vacancy and application / placement life cycles. Use in: SRS 4.4; design 5.1.



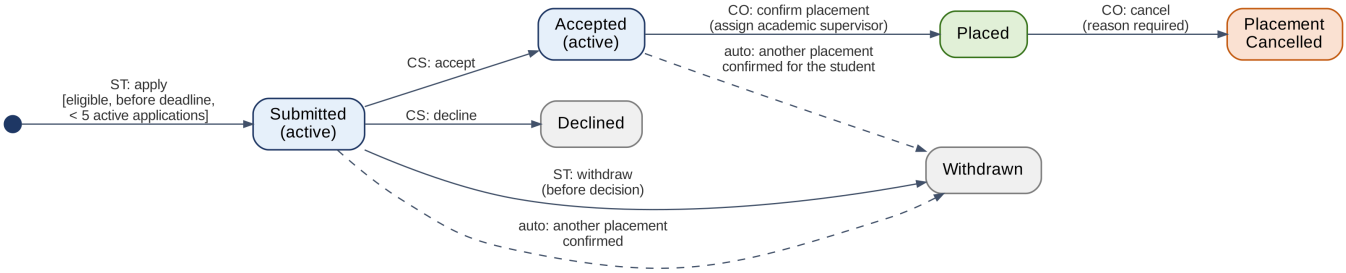


Figure 5 — System architecture

Layers, API modules, AI services, scheduler, data. Use in: Design 2.1, presentation.

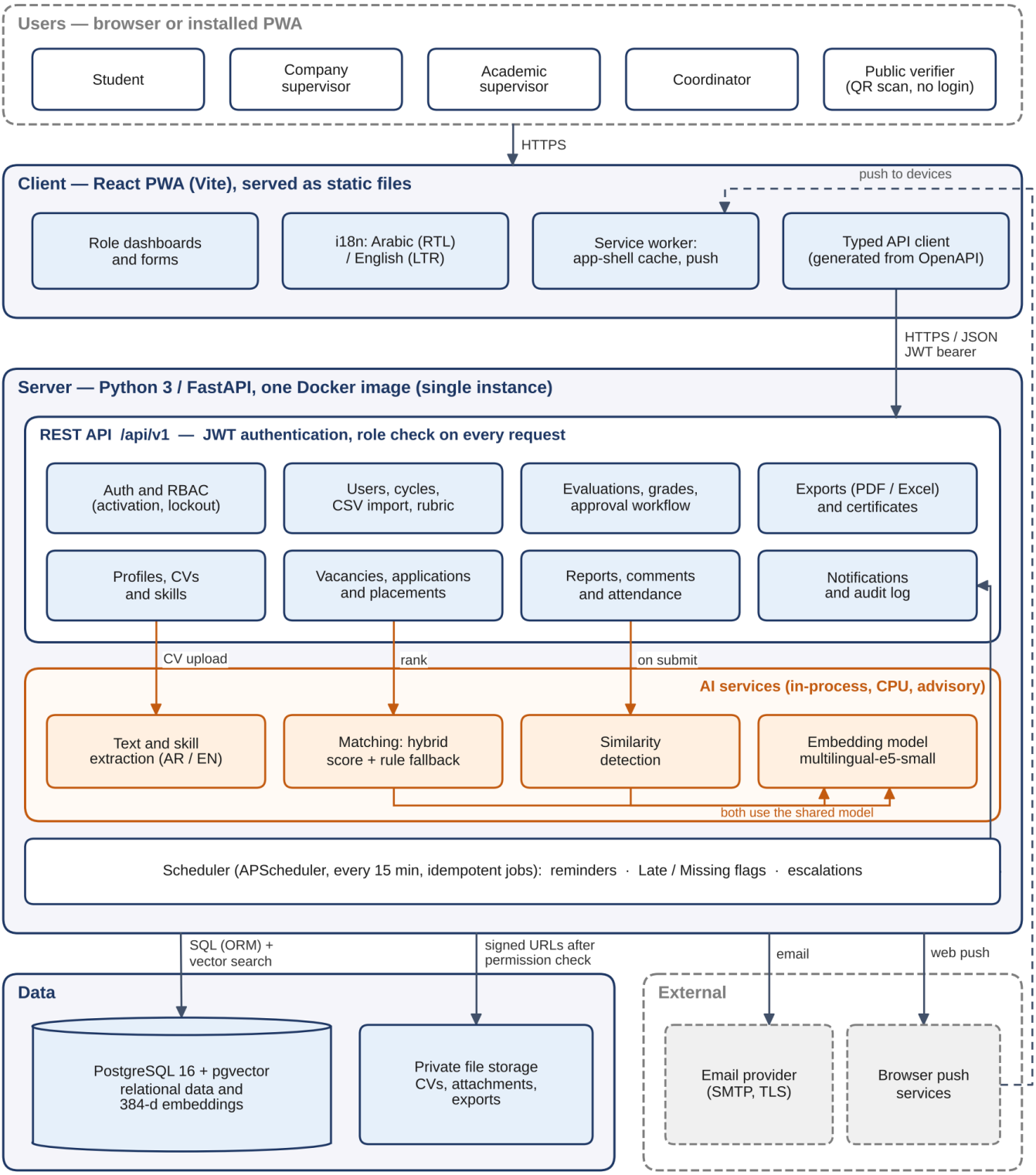


Figure 6a — ERD — accounts, configuration, profiles, vacancies, matching

Tables, keys and cardinalities (crow's foot). Use in: Design 4.2.

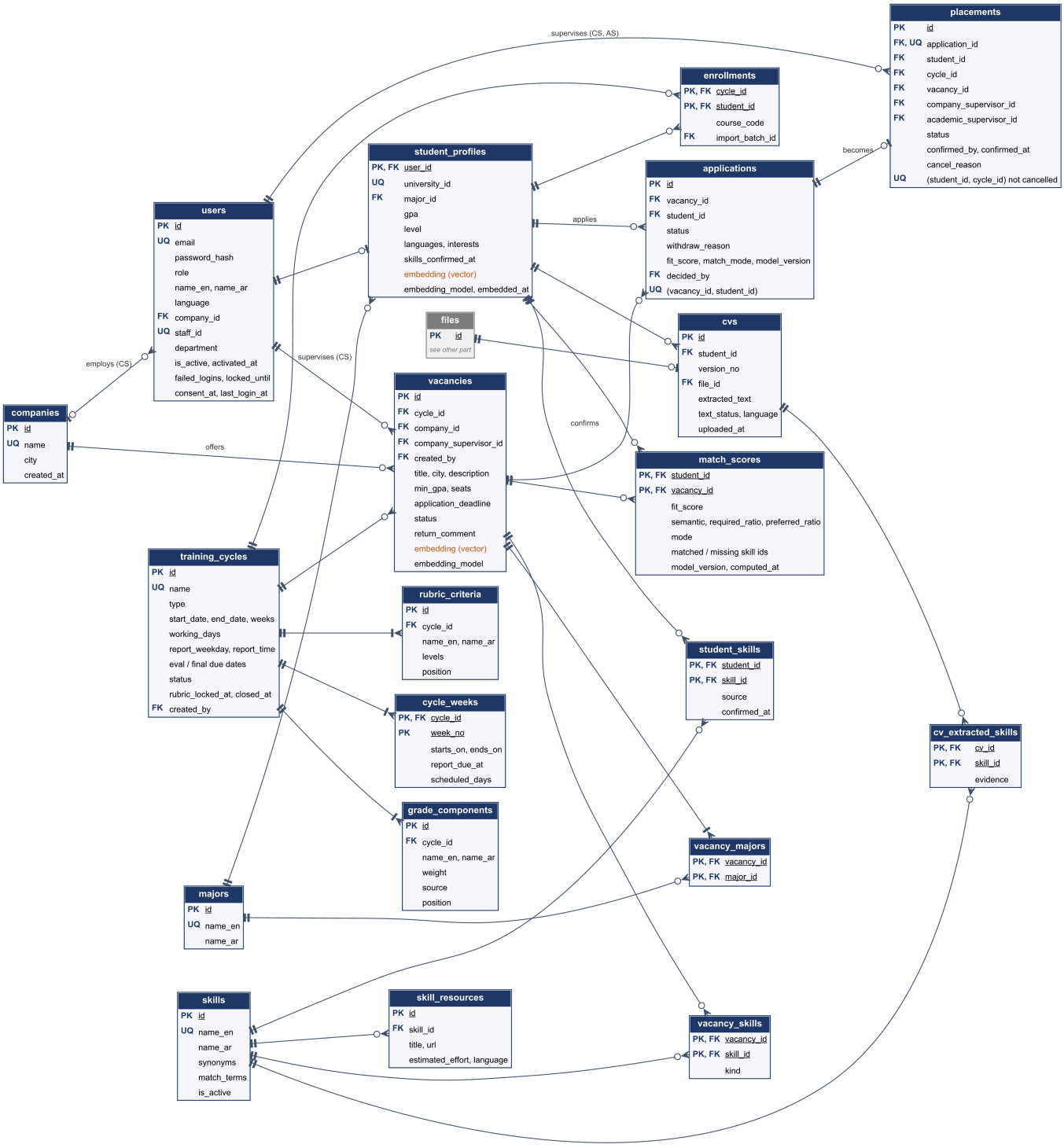


Figure 6b — ERD — reports, similarity, attendance, evaluations, grades

Tables, keys and cardinalities (crow's foot). Use in: Design 4.2.

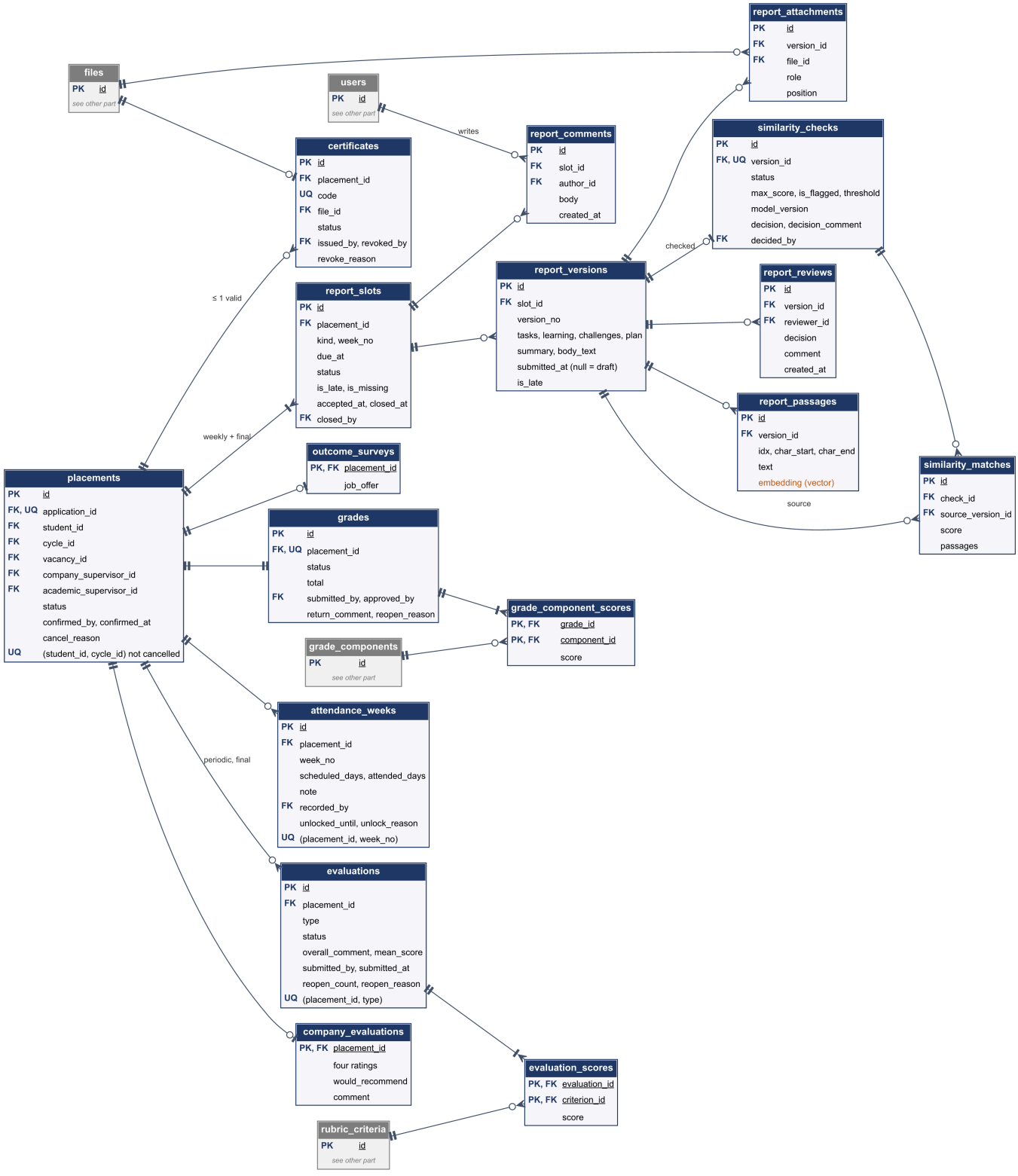


Figure 6c — ERD — platform services

Sessions, tokens, files, notifications, audit, imports, settings, jobs. Use in: Design 4.2.

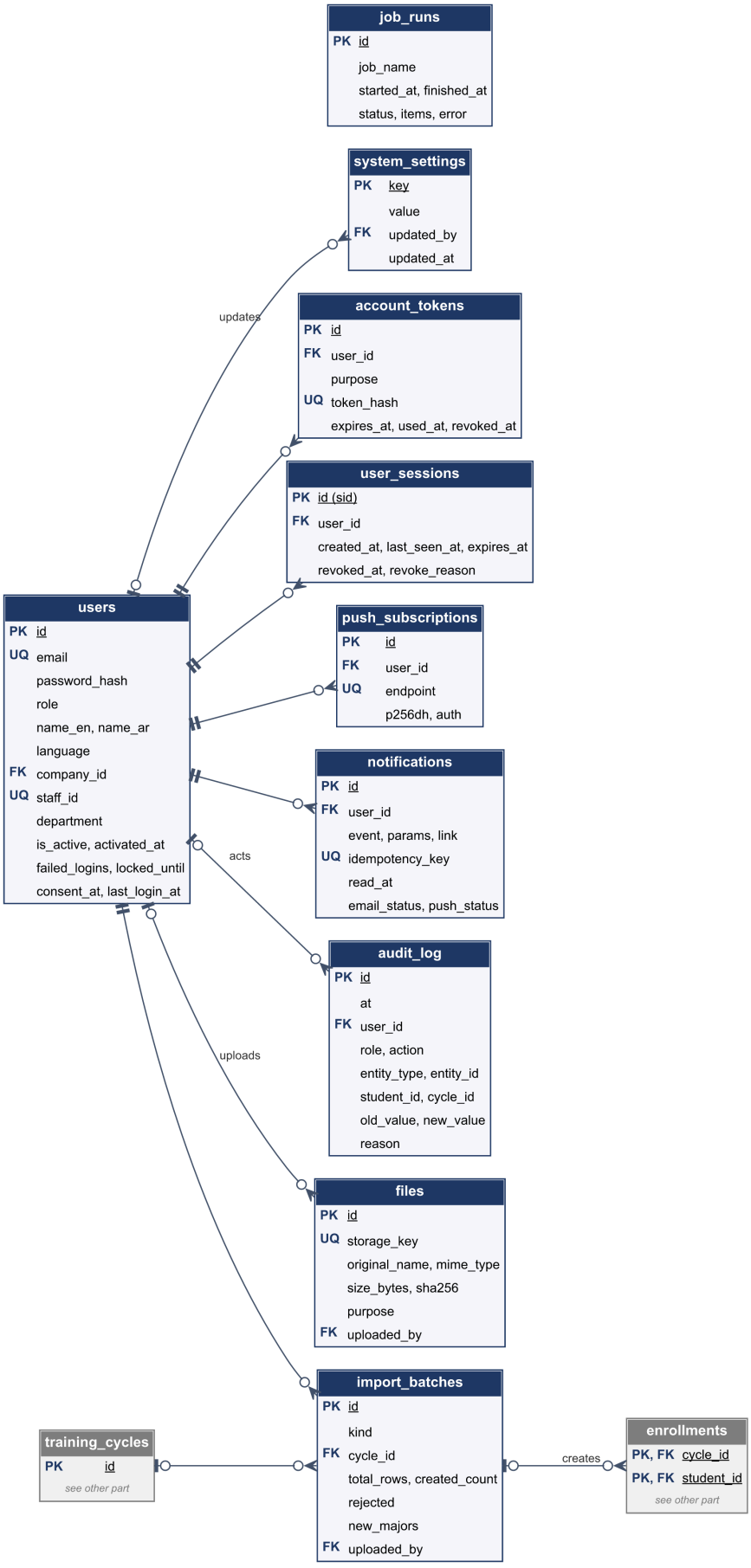


Figure 7a — Sequence — CV upload and matching

Skill extraction, confirmation, ranking with fallback. Use in: Design 6.

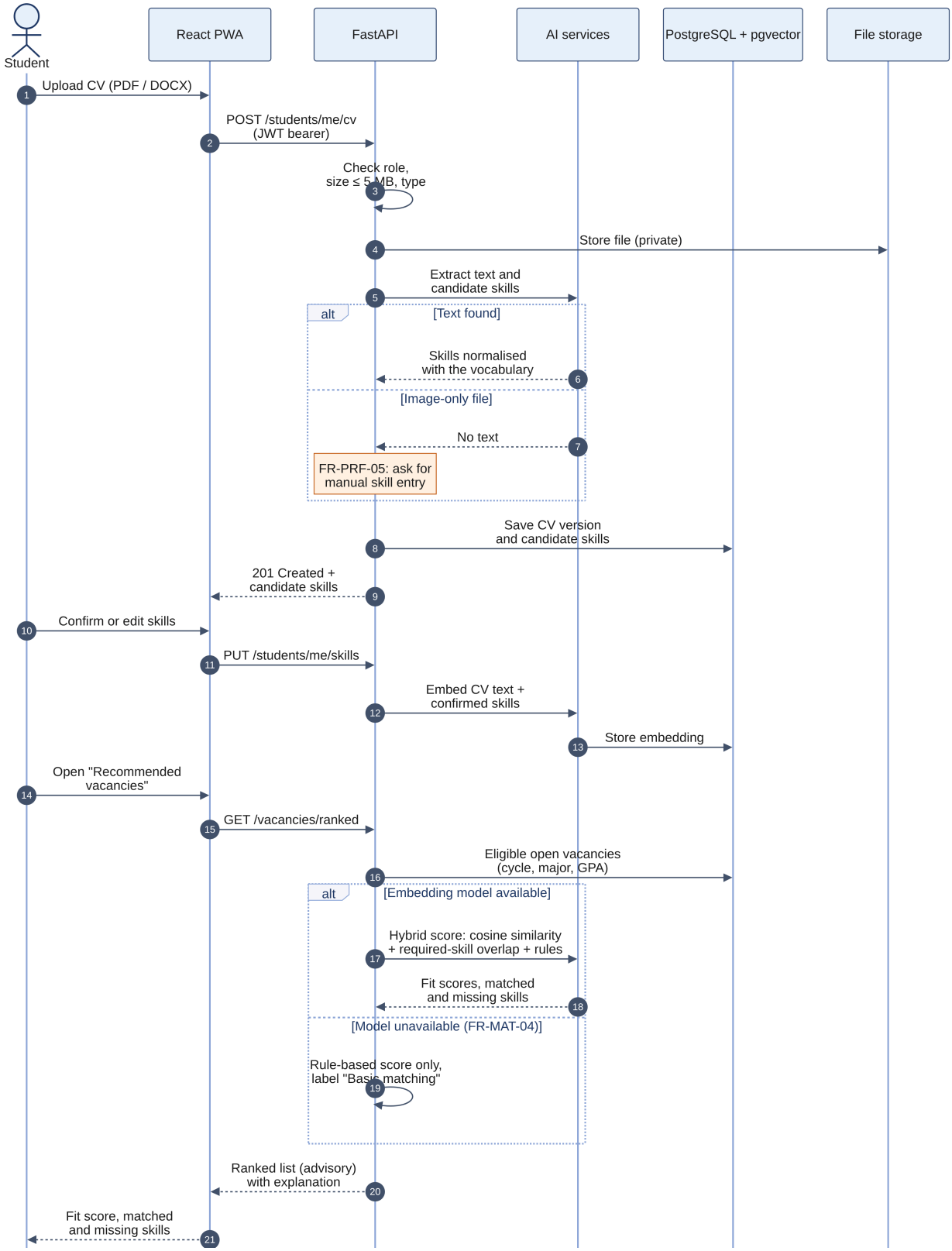


Figure 7b — Sequence — weekly report and review

Submission, Late flag, similarity check, accept / revision. Use in: Design 5.1.

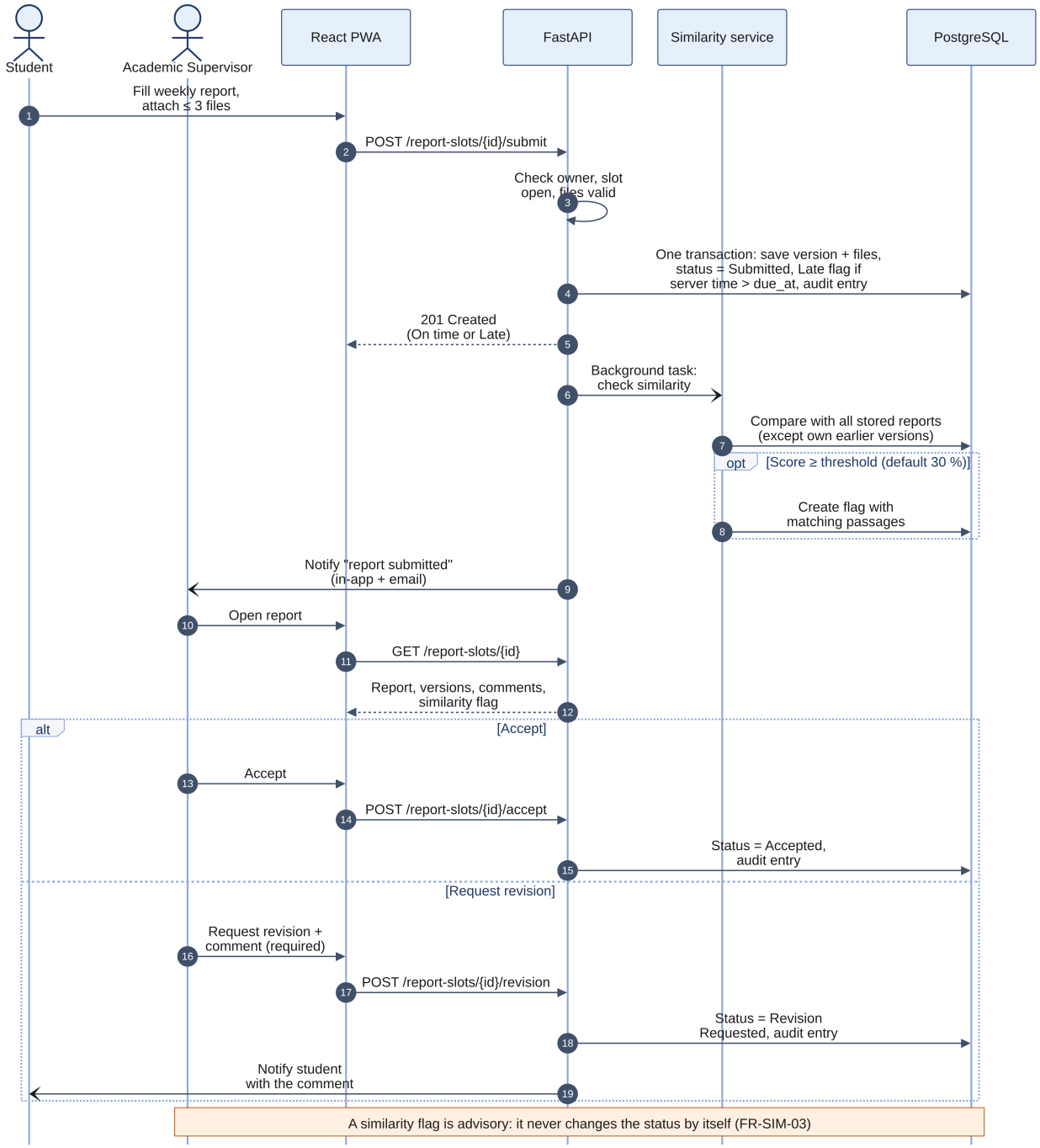


Figure 7c — Sequence — grade submission and approval

Preconditions, approve / return, notifications. Use in: Design 5.1.

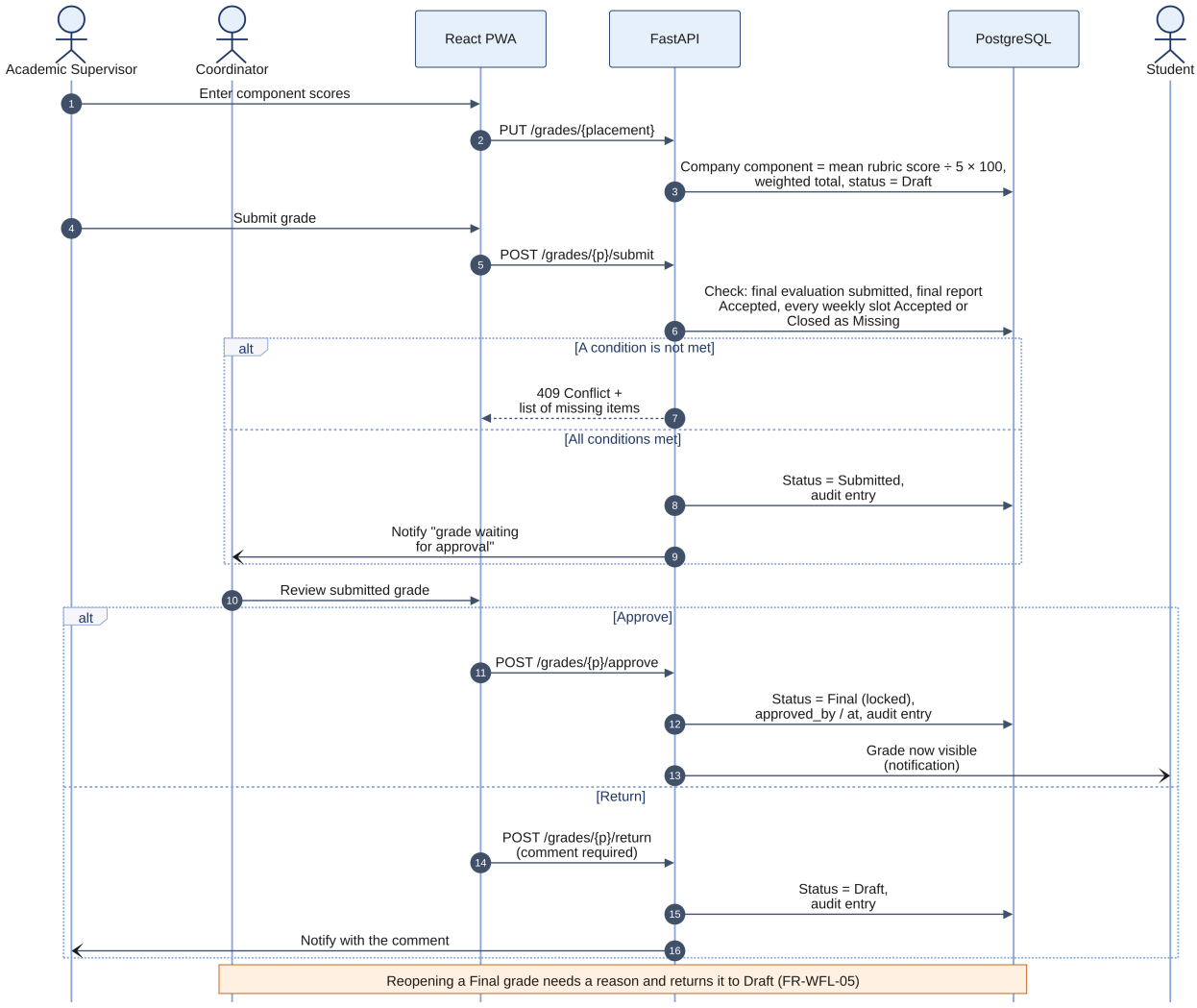


Figure 7d — Sequence — scheduled reminders and escalation

15-minute jobs, idempotent notifications. Use in: Design 5.3.

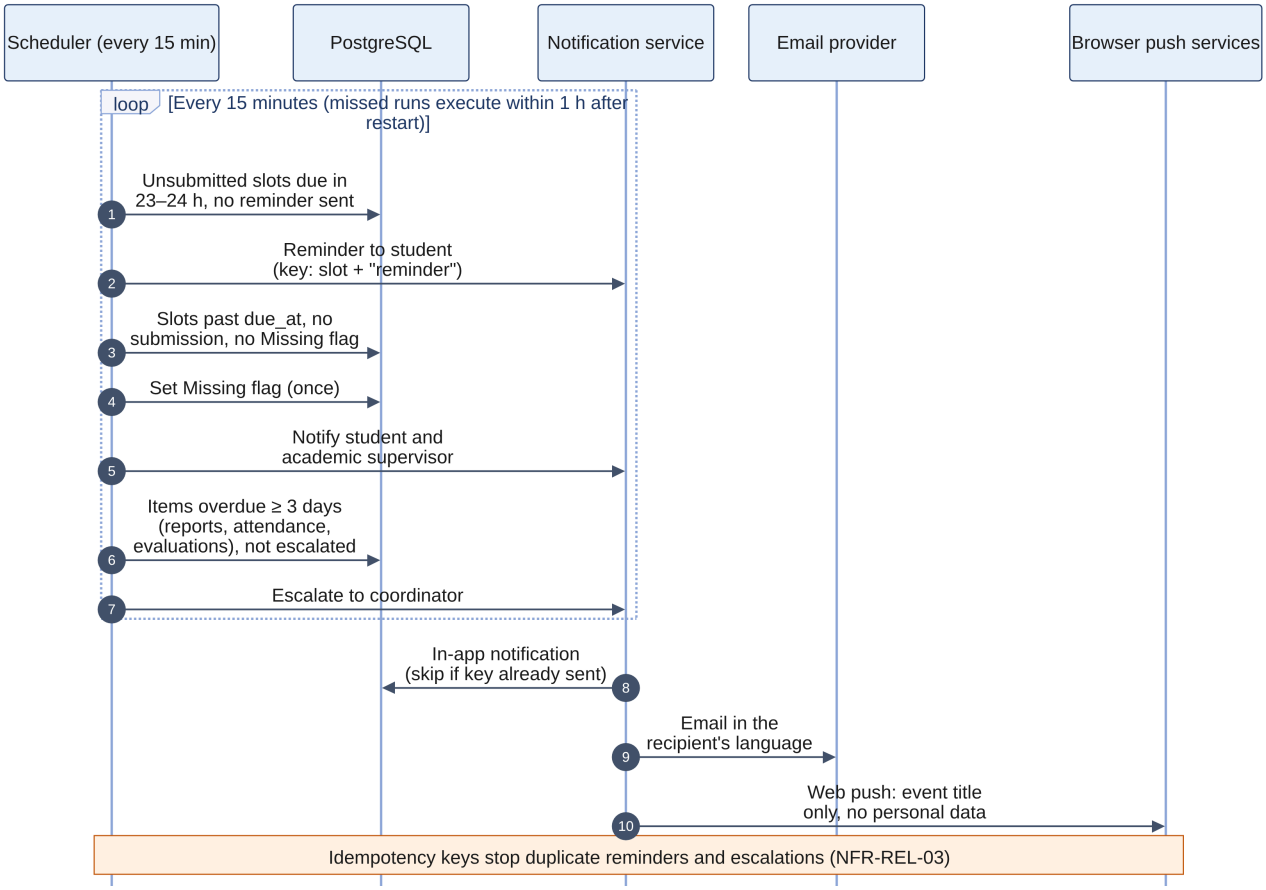
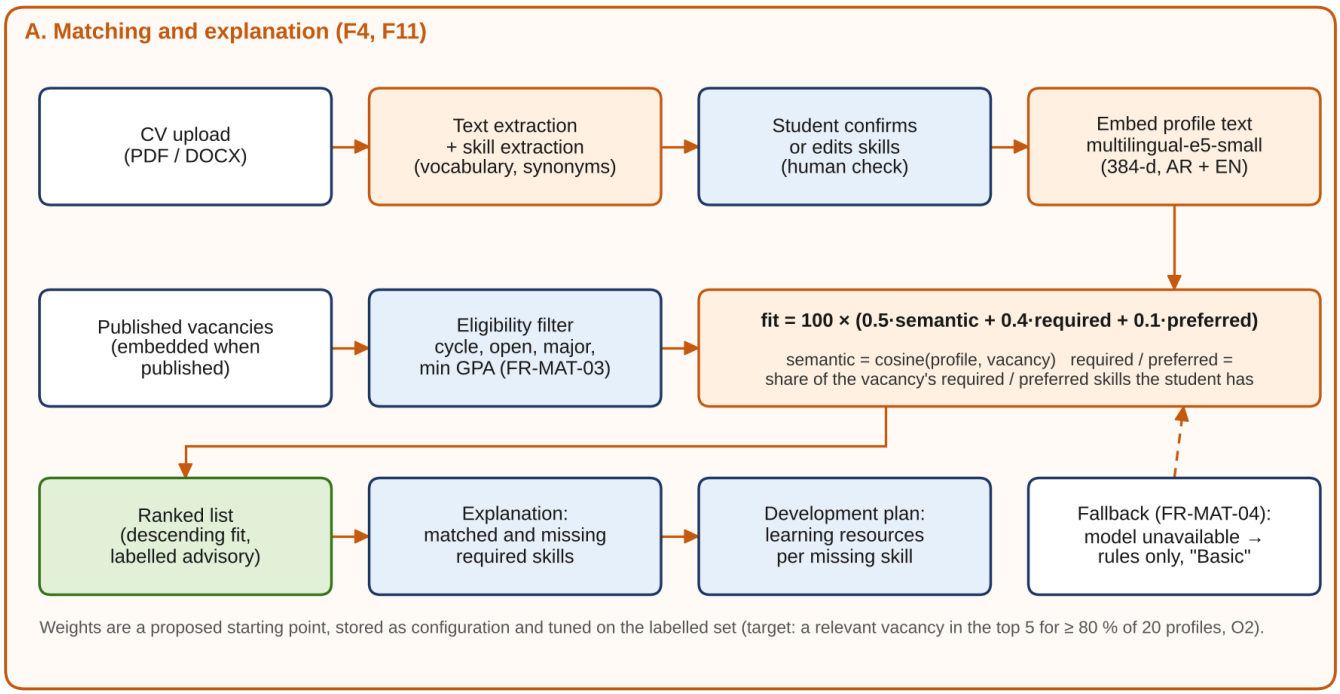


Figure 8 — AI pipelines

Fit-score formula, explanation, similarity method. Use in: Design 6, presentation.



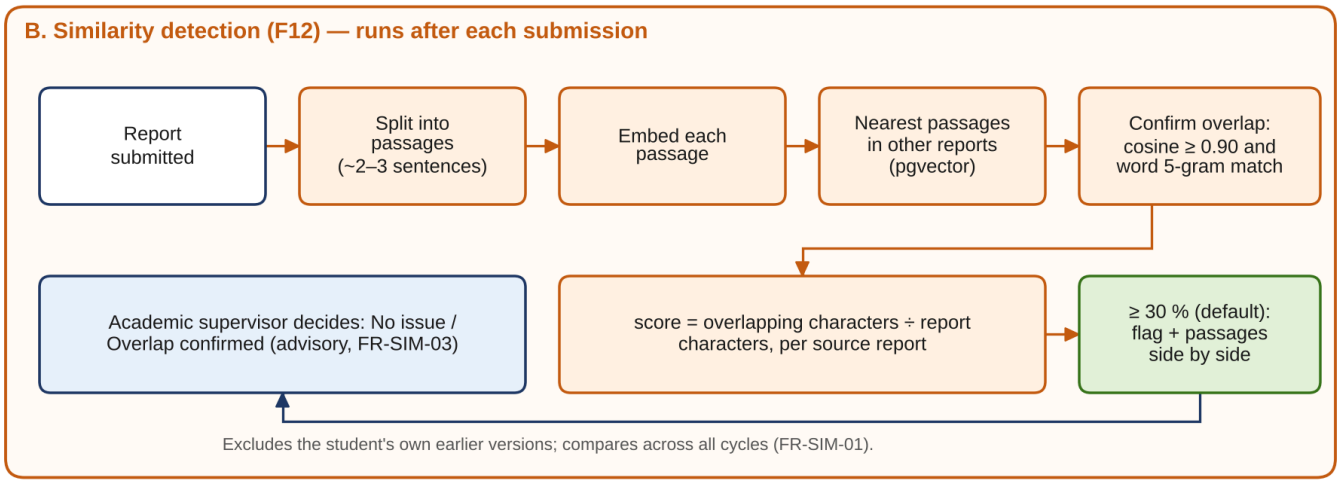


Figure 9 — Deployment

Zero-cost reference deployment (chosen, CR-01) and the self-hosted option. Use in: Design 3.2.

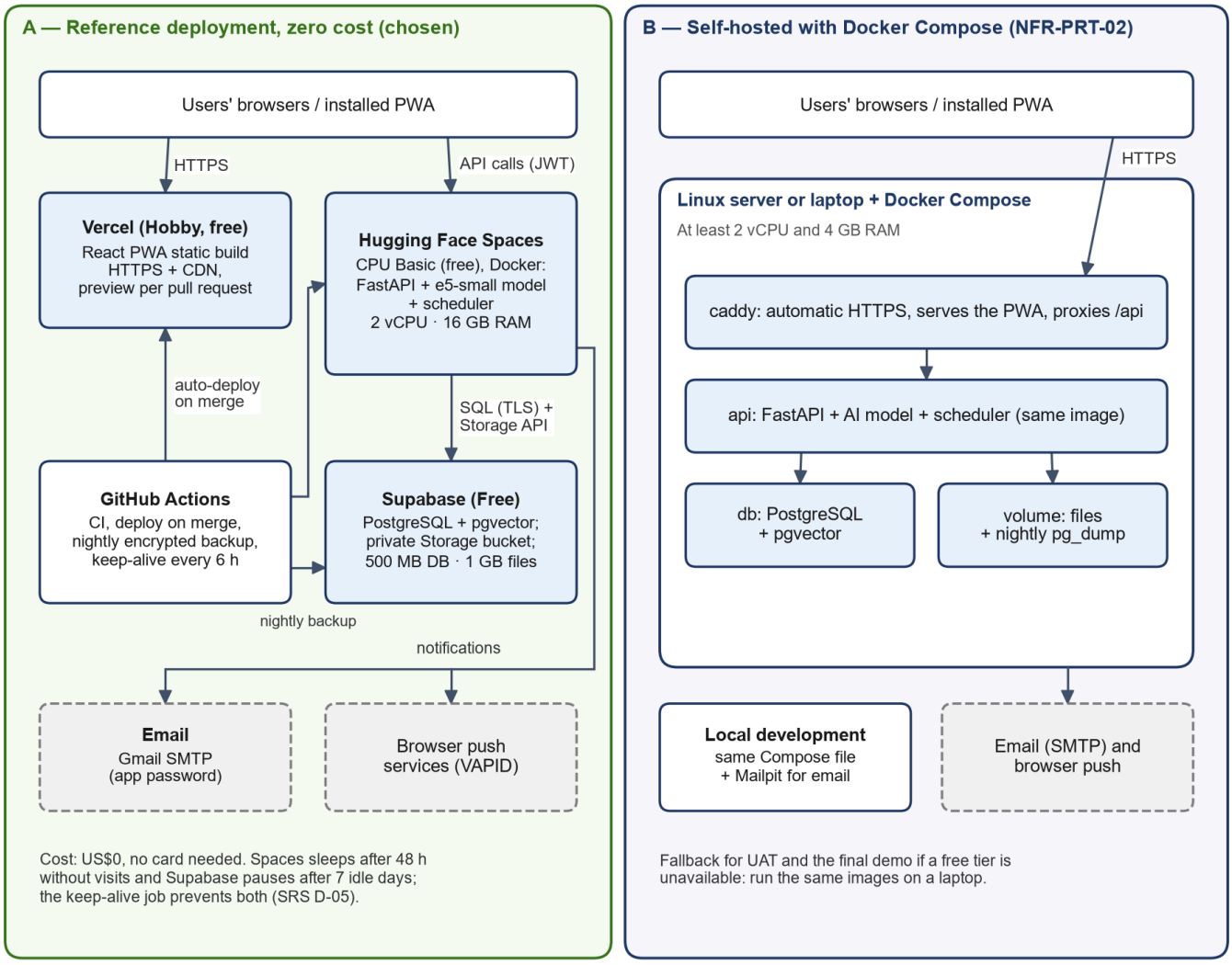


Figure 10 — Schedule (Gantt)

Activities T1–T9, milestones M1–M7, dependencies. Use in: SPMP 4, progress presentations.

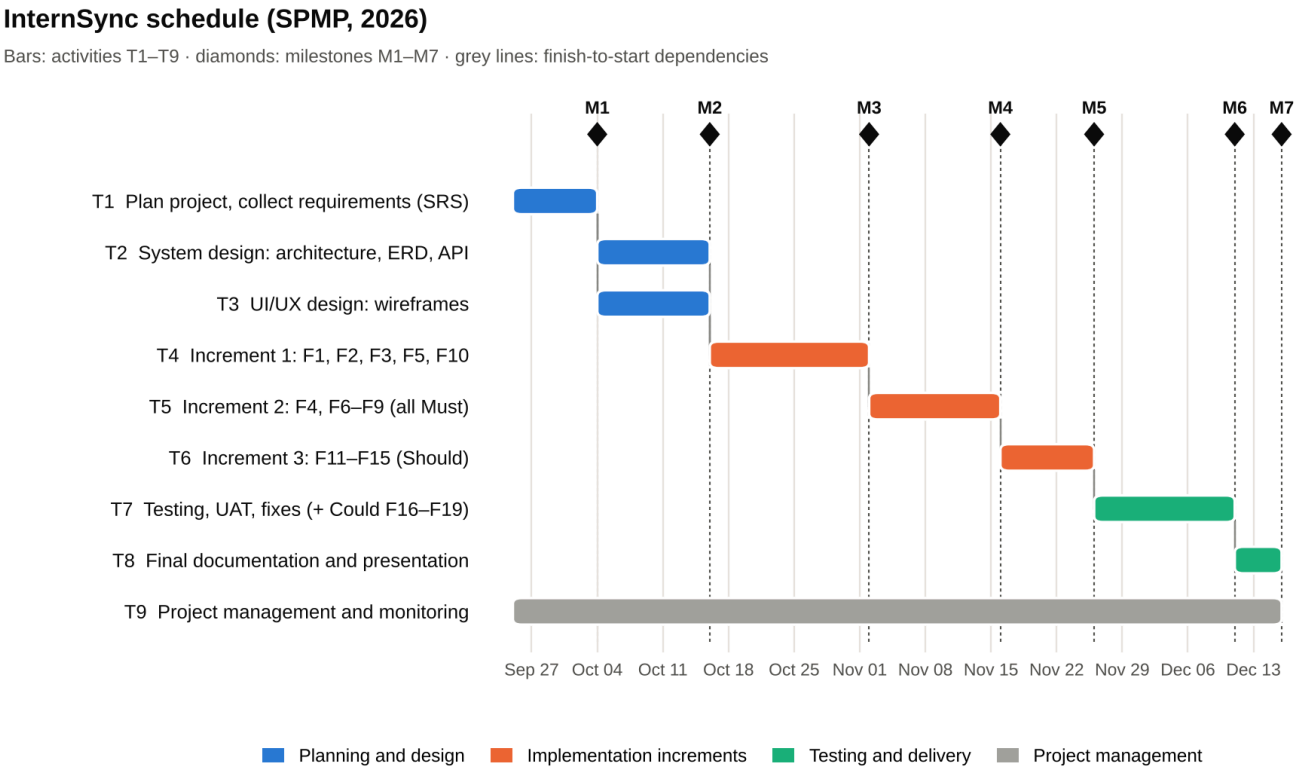


Figure 11 — Requirements by area and priority

116 requirements by area: Must / Should / Could. Use in: SRS 9.2.

